



Ramrao Adik Institute of Technology
Department of Computer Science and Business Systems

NLP Mini Project Presentation

on

"Marathi News Intelligence Pipeline"

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Introduction

- Marathi is spoken by more than 83 million people, yet remains low-resource for NLP
- News outlets publish continuously — BBC Marathi, Lokmat, ABP Majha, Loksatta — but the search, summarisation and classification tools built for English don't carry over
- Marathi is agglutinative: one root takes many suffixed forms — घर → घरात → घरातून → घरांमधून → घराच्या
- A whitespace tokeniser treats all five forms as unrelated words

83M+

Marathi speakers worldwide

Introduction — Objectives

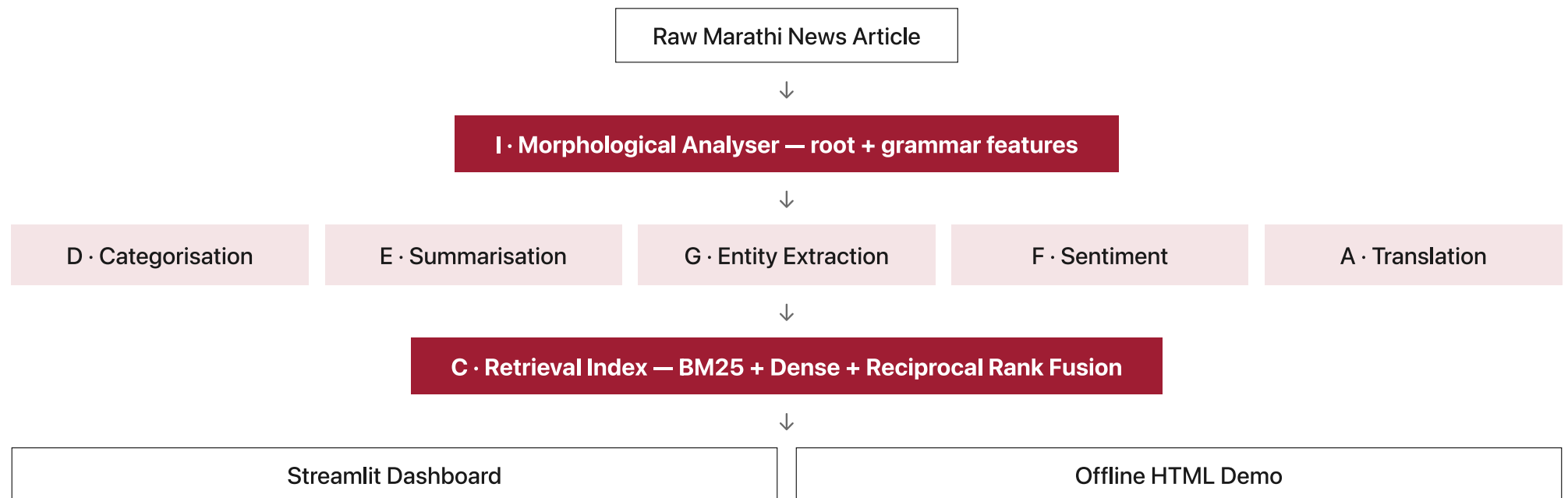
- Build a rule-based Marathi morphological analyser — no training data
- Build a hybrid search engine (BM25 + dense + RRF) and isolate morphology's real effect
- Evaluate every component against a simple baseline, so reported gains are real
- Integrate seven NLP tasks into one pipeline sharing a common representation
- Fine-tune a MahaBERT topic classifier and compare it with a pre-trained baseline
- Ship a usable interface — a Streamlit dashboard and an offline HTML demo

Background / Basic Concept

Surface Form	Segmentation	English Gloss
घर	घर	house (bare root)
घरात	घर + आत	in the house
घरातून	घर + आतून	from inside the house
घरांमधून	घर + ां + मधून	from inside the houses
घराच्या	घर + आ + च्या	of the house

- The analyser strips the suffix chain in four ordered layers — clitic, postposition, verb ending, then case & number — then repairs the stem
- Length is measured in aksharas (syllables), not Unicode characters — fixes cases like शाळा / शाळेत reducing to different roots

Working Principle / Methodology



Results

7

NLP Tasks

3,000

Articles Processed

54.7%

Smaller Search Index

95.4%

Classifier Accuracy

44

Automated Tests

Results — Does Morphology Help Search?

Search Method	Found 1st	Top 5	Top 10	MRR@10
BM25 — plain words	0.600	0.770	0.817	0.675
BM25 + morphology	0.623	0.810	0.830	0.702

न्यायालय (court)

1 result → **14 results**

Same query, same 400 articles, without vs. with morphology

- On one- and two-word queries, morphology gives no ranking benefit
- Its real value is recall: empty result pages fall from 14.4% to 1.1% on cross-inflection queries

Applications

- Search and archival tools for Marathi-language news publishers
- Media monitoring and analytics for brands and PR teams covering Marathi audiences
- Digital libraries and government portals serving Marathi-speaking users
- English-language access to Marathi news, via automatic translation
- A morphology-first approach transferable to other agglutinative Indian languages, such as Hindi and Konkani

Advantages

- No training data needed for the morphological analyser — rule-based from day one
- Every claimed gain is ablation-tested against a baseline, not assumed
- Ships with an offline, single-file HTML demo — no installation needed
- Modular pipeline — each of the seven tasks runs, and fails, independently
- Runs on a 4GB laptop GPU, or CPU only
- Covered by 44 automated tests

Limitations / Challenges

- Morphology's ranking gain is modest, and slightly negative on one- and two-word queries
- Over-stemming: नदी and नद्या both reduce to नद — a consistent label, not a real word
- Entity boundaries can merge two adjacent names of the same type
- The summariser selects existing sentences and does not beat a simple lead-2 baseline on ROUGE
- All articles come from BBC Marathi, so topic mix and tone are skewed

Recent Developments / Future Scope

- Live ingestion already works: pulls current articles from BBC Marathi, ABP Majha and Lokmat
- Add a dictionary check to the analyser so roots are valid lemmas, reducing over-stemming
- Move to abstractive summarisation (e.g. IndicBART) as more GPU memory becomes available
- Train a NER model with proper entity boundaries and add coreference resolution
- Broaden the corpus beyond BBC Marathi to reduce topic and tone skew

Conclusion

Seven NLP tasks, integrated around one morphological analyser built entirely from scratch.

Its real contribution is recall, not ranking — empty search results fall from 14.4% to 1.1%.

54.7%

Smaller search index

95.4%

Classifier accuracy

44

Automated tests passing

References

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THANK YOU